성균관대학교

RESEARCH PORTFOLIO

Closed-Loop Control Systems Robust to Objective and Environment Shift

Resource management and other closed-loop control that adapt through RL policy transfer and LLM agents, built and validated through systems engineering

① **Reinforcement Learning** — Resource Management & Cloud Robotics
② **Agentic AI** — LLM Agents for System Operation (Robotics & V2G)
③ **Systems Engineering** — Simulators, Testbeds, Protocols & Deployment

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yslee.gs@gmail.com · Google Scholar: scholar.google.com/citations?user=GhwtxtgAAAAJ

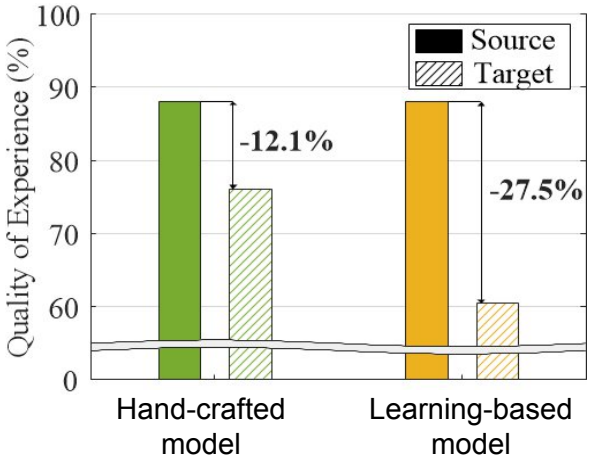
Youngseok Lee — Ph.D. Candidate, Sungkyunkwan Univ.

- **Education** — Integrated M.S.–Ph.D., Electrical & Computer Engineering, Sungkyunkwan University (Mar. 2013 – Feb. 2027, expected) · Advisor: Prof. Honguk Woo (CSI-Agent Group) · Dissertation: Reinforcement Learning for Quality-Driven Network Control
  - 2018: moved to Prof. Honguk Woo's group and defined the dissertation topic · Expert Research Personnel service (Mar. 2017 – Mar. 2020) · Global Ph.D. Fellowship, 5-year term (2013–2018, Ministry of Education / NRF Korea)
- **Achievements** — 8 granted patents (US 5 / KR 3, first inventor on 4) · 5 systems delivered · IEEE INFOCOM 2027 (first author, under review) · NeurIPS 2024 (second author) · 3 SCIE journal papers (first author 2 / second author 1) · 35 earlier publications (2012–2018)
  - Delivered to Samsung Electronics ×2, Polaris AI, Acryl, and ADD (Agency for Defense Development), each as technical leader · 20 funded projects (principal investigator 1 · technical leader 10 · researcher 9) · 9 software copyrights

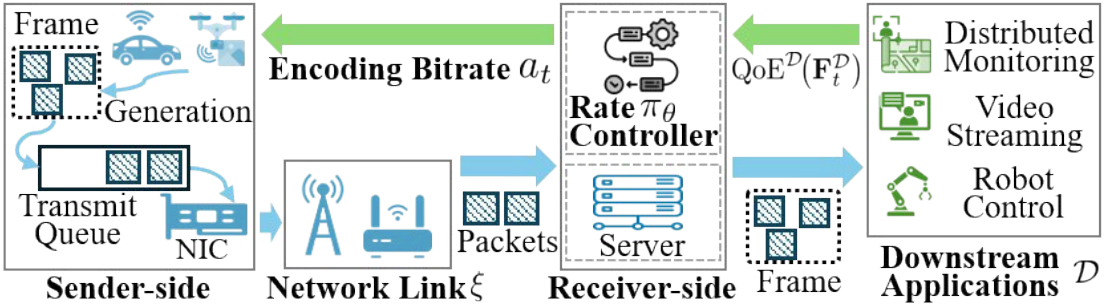
Research area	Representative work	Outcomes
① **Reinforcement Learning** — Resource Management & Cloud Robotics	○ Repot — Bandwidth allocation transferred across network environments by flow embedding and action shaping ○ XferQoE — One QoS-pretrained rate controller, adapted per application by objective-weight fitting from QoE feedback ○ Physical-AI PoC — Code-as-Policies demo · robot foundation model (RFM) fine-tuning on a real arm (SO-101) ○ Off-board RFMs under network delay (in progress; page 12)	○ IEEE Access 2021 (first author) ○ 1 first-inventor patent family (KR 2023 · US 2026) ○ IEEE INFOCOM 2027 (first author, under review)
② **Agentic AI** — LLM Agents for System Operation (Robotics & V2G)	○ LDuS — LLM-guided skill diffusion for zero-shot policy adaptation ○ AgentV2G (in progress) — LLM-agent V2G (vehicle-to-grid) fleet operation under drift, day-ahead → real-time; SAC dispatch policy	○ NeurIPS 2024 (second author)
③ **Systems Engineering** — Simulators, Testbeds, Protocols & Deployment	○ S-Sim packet simulator → NS-3 → 16-node Raspberry Pi NiL testbed ○ Code-as-Policies demo pipeline · SO-101 real-arm fine-tuning setup ○ V2G fleet-operation environment (market rules · SMP · EV sessions) ○ FL with knowledge distillation → edge-NPU YOLOv8n inference pipeline ○ NVMe-oF storage simulator with dual-output DNN surrogate ○ Delay-injection evaluation harness for off-board RFMs	○ Systems delivered ×5 (Samsung ×2, Polaris AI, Acryl, ADD) ○ 3 US patents with Samsung (storage ×2 · IoT resource identification)

Control Robust to Objective and Environment Shift

- **Problem** — A learned model matches a hand-crafted model in the source environment, yet degrades far more severely than the hand-crafted model when the objective or the environment shifts even slightly between source and target (left figure)
  - Such shifts occur after deployment as changes in the application objective (the definition of quality of experience, or QoE) or the execution environment (network condition, demand pattern, market rule); the hand-crafted model degrades gracefully but requires manual re-tuning for each condition, whereas the learned model needs data collection and retraining for each condition
- **Goal** — Closed-loop control that combines the robustness of the hand-crafted model with the performance of the learned model and, when the objective or the environment shifts after deployment, adapts from a small amount of data without retraining the deployed policy
  - ① **Reinforcement Learning** — RL policies for resource management and cloud robotics: a meta-RL rate controller and a bandwidth-allocation policy that transfer to new objectives and environments with little or no retraining (XferQoE · Repot), and a SAC dispatch policy for V2G fleets (AgentV2G), with RL serving control for off-board robot foundation models in progress (page 12)
  - ② **Agentic AI** — AgentV2G uses LLM agents to turn market rules into solver-ready objectives and constraints for V2G operation; LDuS uses an LLM to turn user requirements into executable loss-function code and validate code and trajectories in a feedback loop for skill adaptation
  - ③ **Systems Engineering** — The two approaches are implemented and validated on purpose-built simulators, surrogate models, and testbeds; five systems built with the same engineering were delivered to Samsung Electronics ×2, Polaris AI, Acryl, and ADD
- **Representative system** — QoE-driven control loop (right figure): RL-based rate controller → network link → application QoE feedback



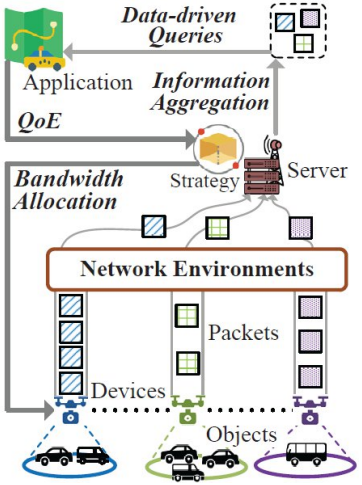
Motivating measurement (Repot)



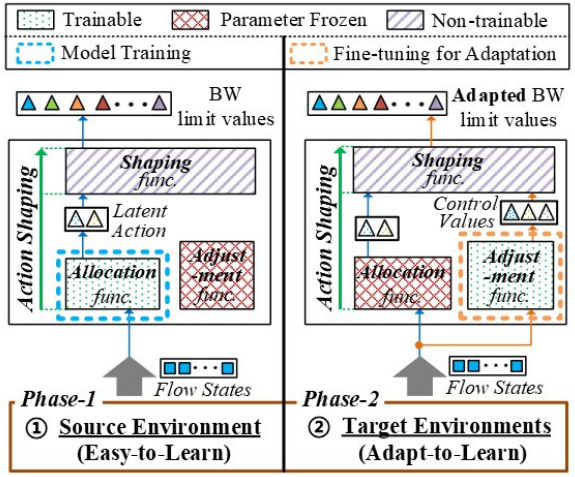
Closed-loop QoE-driven adaptive rate control (XferQoE)

[RL, Finetuning] Repot — Transferable RL for Networked Monitoring

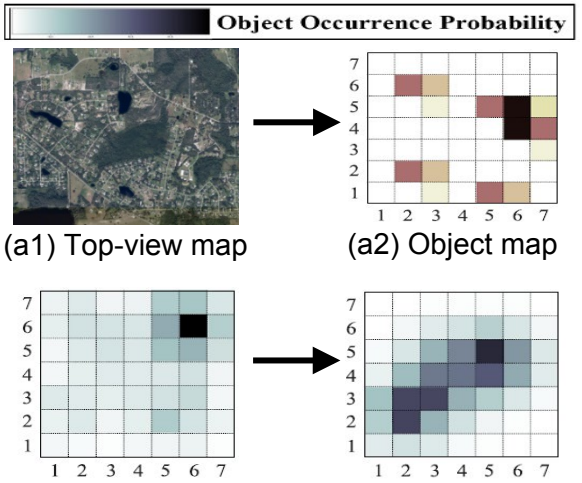
- **Overview** — 2020–2021 · IEEE Access 2021 (first author) · Network Monitoring System patent, KR 2023 / US 2026 (first inventor)
- **Role: Leader** — problem definition, algorithm design, implementation, experiments
- **Problem** — When sensing devices share a single link, the bandwidth allocation must keep a server's continual query accurate as the network degrades. Re-learning that allocation from scratch for each new network scale and condition is prohibitively expensive (left figure)
- **Approach** — Two-phase transfer (center figure): ① the policy is trained end to end with SAC (soft actor–critic) in an ideal time-slotted source environment with inexpensive samples; ② in the NS-3 target, all learned modules are frozen and a second, small SAC policy (the adjustment function, with its own critics) is trained on expensive samples. Its outputs enter the shaping function together with the frozen latent action
- **Structure** — Flow embedding: a Transformer encoder (per-device MLP vectorization + self-attention over all flows) represents every stream by its history and its relation to the others in one vector space → Allocation: a trainable function emits a latent action → Shaping: a non-trainable function maps it to per-device bandwidth limits by a distance-weighted softmax; the adjustment function supplies the control values
- **Result** — The policy adapts with 6.25% of the from-scratch training samples and reaches 14.5–20.8% higher QoE (quality of experience) than the two learning-based baselines in the target. QoE holds at 91–94% of the equal-share reference across 49–400 devices in the source, and the SUMO (Simulation of Urban MObility) vehicle study shows +6.5% (urban) and +12.6% (suburban) over Top-Opt, a Top-K heuristic (right figure)



Networked monitoring systems



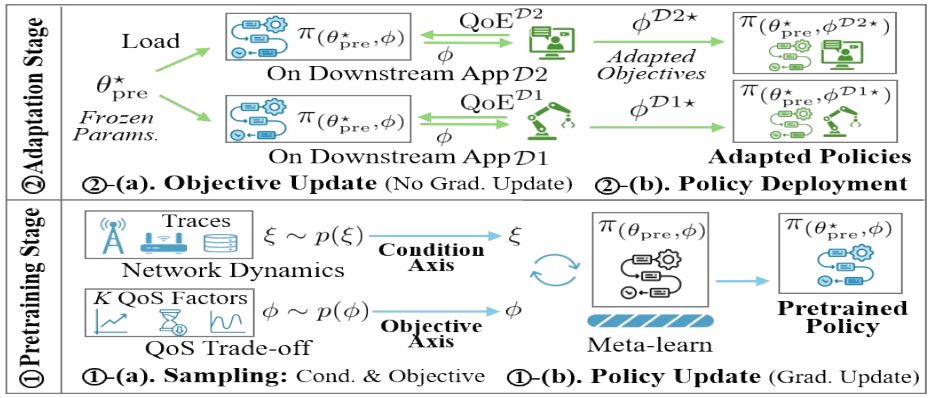
Action shaping model structure



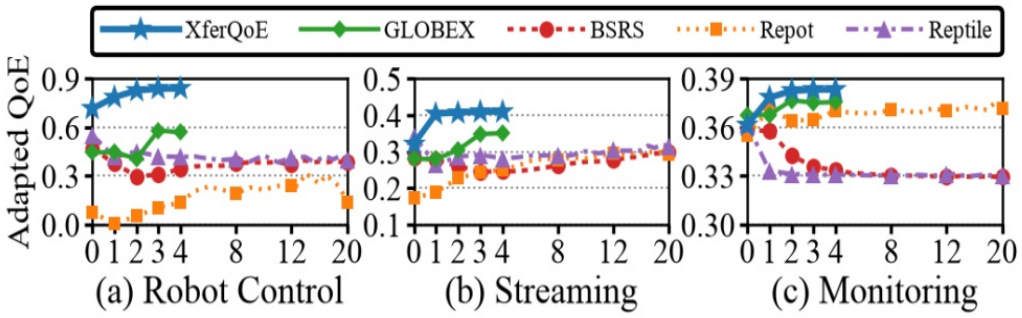
Action attention map transfer

[Meta-RL, Transformer] XferQoE — QoS-Pretrained Rate Control for Model Serving

- **Overview** — 2024–2026 · IEEE INFOCOM 2027 (first author, under review) · Physical-AI Core-Technology PoC, MSIT (Ministry of Science and ICT) / NIPA (National IT Industry Promotion Agency), 2025
- **Role: Leader** — (full cover) problem definition, algorithm design, implementation, experiments, testbed
- **Target system** — A cloud robot streams observation frames over a wireless link to an edge server that runs the manipulation policy, so the served model receives only the frames the link can carry. On a limited link the highest frame quality is not the highest QoE (quality of experience), because quality gained at the cost of timeliness deprives the control loop of up-to-date observations. The best stream differs per deployment and the served model (e.g., a robot foundation model, an LLM, or a recognition model) cannot be fine-tuned for each, so XferQoE leaves the model untouched and controls the sending rate, and with it the quality, of the remotely sourced input
- **Problem** — Each application weighs the QoS (quality of service) factors (timeliness, throughput, smoothness) in its own way, and its QoE is observable only through the complete network-to-application stack, so training a controller per application incurs both costs again: the per-application objective design and the full-stack QoE collection
- **Approach** — Meta-RL (meta-reinforcement learning) in two stages: ① one rate controller is meta-pretrained on QoS objectives alone in S-Sim (Simplified Simulator), each episode sampling a task over two axes, the bandwidth dynamics (condition) and the objective weight w (trade-off), so the policy learns to infer the condition from recent history instead of being retrained for it; ② the policy is then frozen and only w is fitted from a few QoE feedback shots on each downstream application, with no gradient update (left figure)
- **Structure** — SAC (soft actor–critic) with one module per meta-training axis: a regime–context encoder (a bidirectional Transformer with mean pooling → regime latent; a per-patch 1-D CNN with a cross-attention readout → step latent, 8-dim each) and factor critics (one Q-head per QoS factor, ensemble of 3) that compose the value linearly under w. The actor reads the state, both latents, and w (policy ≈ 0.3 MB, encoder ≤ 3.5 MB)



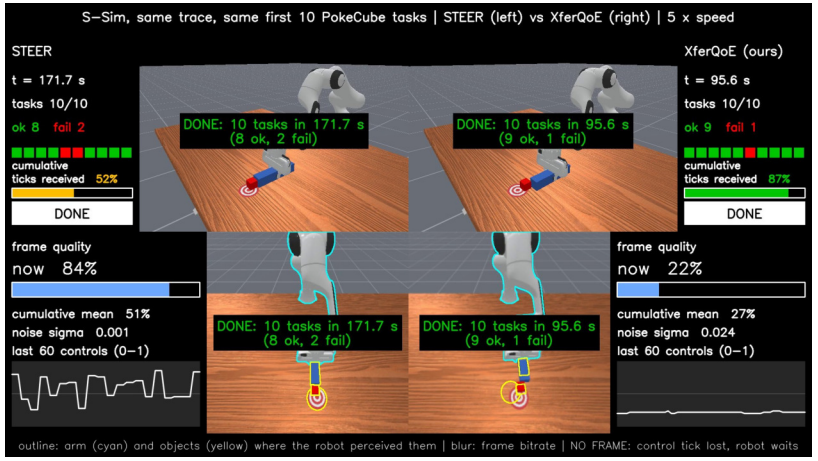
QoS pretraining → QoE adaptation



Adapted QoE per adaptation shot (one shot = one adaptation round)

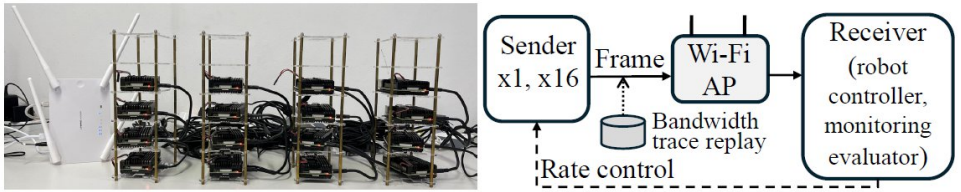
[Sim2Real] XferQoE — Frozen Rate Controller on a Real Wi-Fi Link with a Simulated Robot

- **Target system** — The edge server runs the manipulation policy and the rate controller, and only observations cross the wireless link. The Sim2Real target is therefore the network side, while the robot runs in ManiSkill3 (SAPIEN, learned policy) and Meta-World (MuJoCo, scripted policy) simulation and its task QoE (quality of experience) is defined from manipulation success, control continuity, and completion efficiency
- **Built** — The NiL (network-in-the-loop) testbed: one Raspberry Pi 4 sender for robot control or up to 16 for monitoring on a real 802.11ac link via an access point to the edge server. C/C++ daemons implement episode-level clock synchronization. A sender-side queue model and frame-deadline expiry accounting track traffic; the receiver reconstructs queue growth, dropped bytes, and frame lateness from sender-stamped packet headers
- **Validation path** — Three network environments of increasing fidelity replay the Puffer bandwidth trace (the hardest to predict of the eight traces) behind one state-input, rate-output interface shared by all three: packet-level C++ S-Sim → full-stack NS-3, the NiL testbed, a real Raspberry Pi 4 with its OS stack and packet timing. The same frozen rate controller runs on all three with its parameters unchanged
- **Result** — On the link, only the objective weights are refit while the rate controller stays frozen. XferQoE has the highest QoE in all three real-link evaluations. ManiSkill3 QoE is 0.439 vs. 0.367 for the per-application hand-crafted controller (success rate 67.8% vs. 64.7%, table)
- **Demo video, S-Sim** — Same trace, first 10 PokeCube tasks at 5× speed, one example run: STEER (left, a learning-based rate controller) keeps mean frame quality at 50.6%, but the robot receives a complete observation on only 51.6% of its control ticks and finishes in 171.7 s with 2 of the 10 tasks failing. XferQoE (right) lowers mean frame quality to 27.2%, reaches 86.8% of ticks, and finishes in 95.6 s with 1 failing



Demo video: STEER vs. XferQoE

sonny0714.github.io/demo_video_robot/xferqoe-vs-steer.html



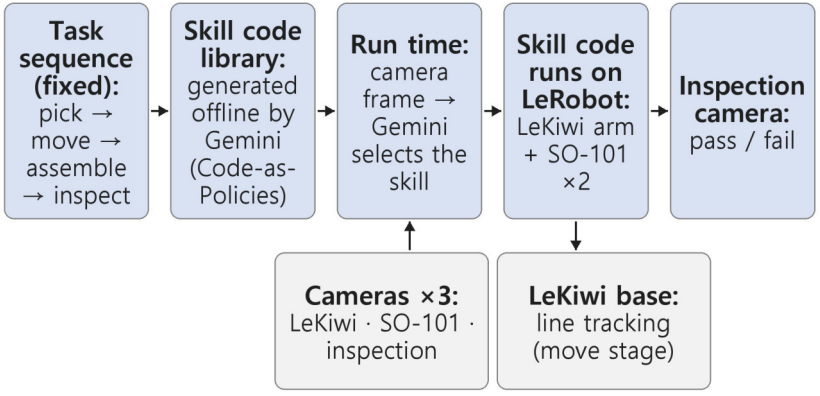
NiL testbed: Raspberry Pi 4 senders, Wi-Fi AP, edge server

Real-link robot control, ManiSkill3 PokeCube (HC = hand-crafted)

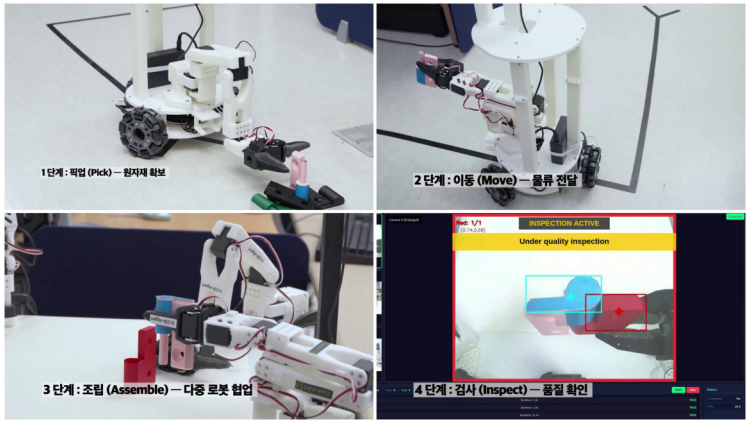
Method	QoE	Success Rate (%)	Frame Quality (%)	Control Continuity (%)	Efficiency (%)	Working Time (s)	Completion Time (s)
XferQoE	0.439	67.8	27.3	88.5	81.4	7.7	9.34
STEER	0.003	54.2	37.5	59.1	86.6	7.2	14.54
GLOBEX	0.203	60.7	33.4	74.0	85.4	7.2	10.94
Per-app HC	0.367	64.7	25.7	86.7	80.8	7.7	9.36

[RFM, VLM, VLA] Physical-AI PoC — Code-as-Policies Demo, RFM Fine-Tuning

- **Overview** — Physical-AI Core-Technology PoC (MSIT / NIPA, 2025, researcher): the four-stage demo was recorded in Jan. 2026. The SO-101 work (Mar.–May 2026) is a separate effort toward human-to-robot transfer
- **Role: Participant** in the PoC demo (pipeline structure design, LeRobot runs; jointly with the lab) and carried out the SO-101 work (encoder pretraining, data collection, fine-tuning, real-arm tests)
- **Demo: skill-based Code-as-Policies** — The task sequence is fixed (pick → move → assemble → inspect) and the skills are predefined. For each skill, the prompt to Gemini (a VLM, vision-language model) carries a code skeleton written over the robot's perception and control primitives, and Gemini fills the skeleton with executable code offline; the filled skills form the library. At run time Gemini reads the camera frame, selects a skill, and that code drives the arms on LeRobot (LeKiwi's arm picks, two SO-101 arms assemble). The LeKiwi mobile base moves by line following and an inspection camera returns pass/fail (video at right). No learned policy is in the loop
- **SO-101: RFM (robot foundation model) encoder fine-tuning toward human-to-robot (H2R) transfer** — The vision encoder was the part trained so that human videos could guide robot imitation learning: H2R was reproduced on public data (two ViT-B/16 encoders pretrained by masked autoencoding, on human and on manipulation videos, each as the ACT (Action Chunking with Transformers) backbone, compared offline with ImageNet pretraining). On the real arm, the vision encoders of the pretrained SmolVLA and π0.5 were then fine-tuned on LeRobot, action expert frozen, on teleoperated donut pick-and-place data (104 episodes, 45,553 frames). ACT on the same data was the baseline
  - **Models** — ACT: CNN vision backbone + transformer encoder–decoder trained as a conditional VAE; SmolVLA: SmolVLM-2 vision-language backbone + flow-matching action expert; π0.5: PaliGemma-based VLM + flow-matching action expert
- **Real-arm result** — On donut pick-and-place, 10 trials each at trained positions (in-distribution, ID) and new positions (out-of-distribution, OOD), reported as [ID, OOD]: ACT [70%, 40%], SmolVLA [90%, 50%], π0.5 [80%, 60%]. The gaps between policies are one or two trials of ten and the training budgets (steps, batch) differ, so no ranking or pretraining effect is claimed



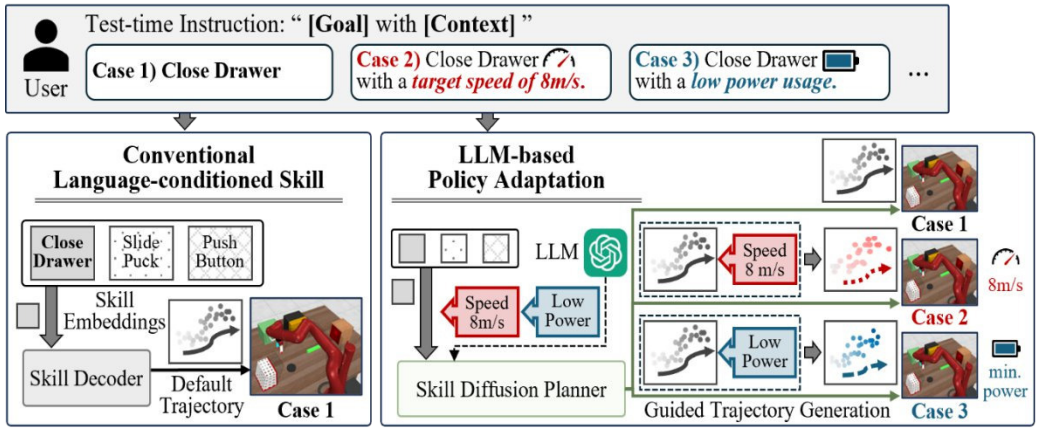
Demo: offline skill code; camera → Gemini selects skill → LeRobot



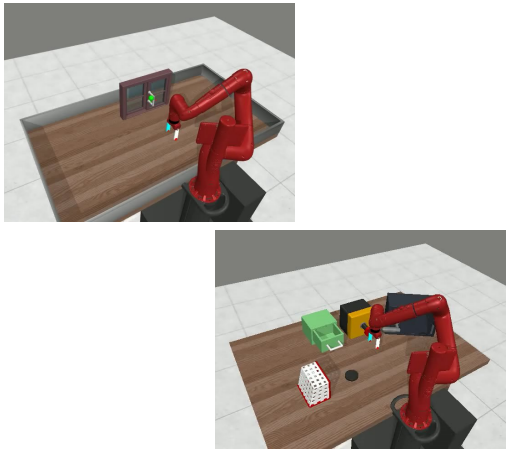
PoC demo video (41 s): pick, move, assemble, inspect

[Diffusion, Action Decoder] LDuS — LLM-Guided Skill Diffusion for Policy Adaptation

- **Overview** — 2023–2024 · "LLM-based Skill Diffusion for Zero-shot Policy Adaptation," NeurIPS 2024 Main Conference Track (second author)
- **Role: Leader** — problem definition, algorithm design, architecture, experiment design; robot-task validation on Meta-World
- **Problem** — A skill-based policy learned from expert data cannot adapt to a language-specified context unseen in training, and retraining for each context is expensive (left figure, Cases 2–3). Such a context is a changed environmental condition or a user requirement such as a target speed, low power usage, or a spatial constraint
- **Pretraining (skill learning)** — On goal-annotated expert trajectories without context, a hierarchical skill model is trained: a VAE skill encoder maps an h-step state–action segment to a skill embedding, a skill prior predicts it from the state and the language goal encoded by CLIP (contrastive language–image pretraining), and a skill diffusion planner, a U-Net denoiser as in Diffuser, reconstructs the segment from the embedding. Sequential in-painting fixes the past states and actions into the generated trajectory, so the planner completes the rest at test time
- **Adaptation (zero-shot, no policy update)** — In the manner of Code-as-Policies (the LLM writes executable code instead of acting directly), but for a loss function rather than the policy, GPT-3.5 turns the context into loss code, and the gradient of that loss guides every denoising step of the pretrained planner. GPT-4 judges contact from the camera frame to gate constraints that apply only while touching the object. A closed loop validates the code and the guided trajectory, regenerating faulty code and raising the guidance frequency until the context is satisfied
- **Result** — LDuS adapts zero-shot to user constraints on how a task is done (speed, force, energy use, spatial bounds) never seen in training, with no policy update. Success rate / context reward vs. Diffuser+Guidance (same gradient guidance, hand-written loss): Meta-World speed constraints (10 goals) 94.6% / 87.4 vs. 76.4% / 69.8; multi-stage Meta-World (3 long-horizon goals) 84.0% / 82.0 vs. 36.7% / 45.9; a force constraint active only while touching the object, contact judged from the camera frame (2 goals, right figure) 93.1% / 63.5 vs. 1.4% / 33.0



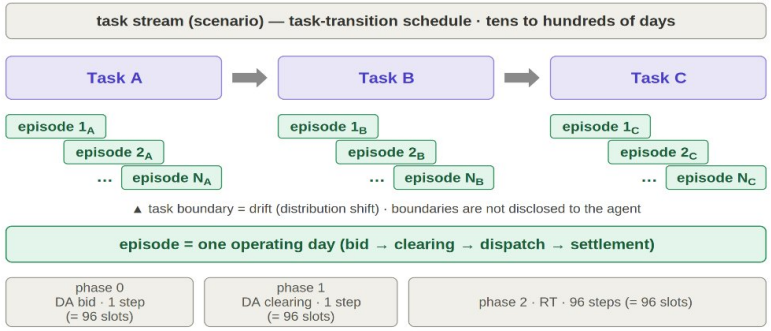
LDuS: language context → LLM-generated loss code → guided trajectory generation



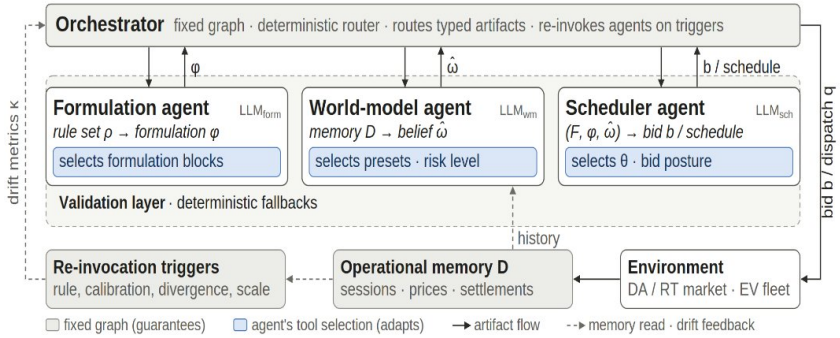
Manipulation task executions (Meta-World)

[RL Solver, LLM] AgentV2G — Agentic V2G Fleet Scheduling Under Drift

- **Overview** — 2026 – present · Hyundai Motor · Kia V2X (vehicle-to-everything) project (Mar. 2026 – Mar. 2027)
- **Role: Leader** — (full cover) problem definition, algorithm design, implementation, experiments; construction of the V2G fleet-operation environment
- **Problem** — A V2G (vehicle-to-grid) charging-hub operator commits irreversible market bids before prices, cleared fractions, and vehicle behavior are known, all drifting over weeks of operation. An agent free to re-plan cannot be trusted to commit, while a fixed pipeline cannot follow the drift
- **Method** — A static orchestration graph with dynamic tool selection (right figure): every operating day follows the same bid → clear → dispatch sequence, so the graph is fixed once, and each agent adapts only by re-selecting among verified tools where drift affects the estimates or rules
  - **Three role agents** — A formulation agent derives the objective and constraints from the market rules and re-derives them when the rules change. A world-model agent maintains a belief state over the unobserved inputs (prices, clearing, sessions) from an operational memory and selects the risk level of its belief corrections. A scheduler agent produces every bid and schedule through an optimization engine (MILP, mixed-integer linear programming)
  - The orchestrator re-invokes an agent when a deterministic drift metric flags a change and validates every emitted artifact (formulation, belief, schedule) through membership, bounds, and scale checks and falls back to deterministic defaults when a check fails. The objective is immutable within an episode, so no agent can relax the objective it is evaluated against
  - **RL solver** — A SAC (soft actor–critic) policy over per-vehicle embeddings and their fleet mean, built to replace MILP in real-time dispatch
- **Evaluation** — Day-ahead and real-time market scenarios on real SMP (system marginal price) and EV (electric vehicle) charging-session data, under Jeju pilot rules with EU and NEM (Australian National Electricity Market) presets; fleets of 1–1,000 EVs; a single open-weight LLM (Llama-3.1-8B-Instruct) without fine-tuning; the RL solver is benchmarked against MILP, rule-based, and oracle dispatch on 200-EV days. Metrics: episode-level operating quality and recovery under price, fleet-behavior, and market-rule drift, individually and combined (left figure)
  - **Scalability** — A 10,000-EV day is scheduled in ≈54 s, within Hyundai Motor's 90 s limit (feedback loop disabled)



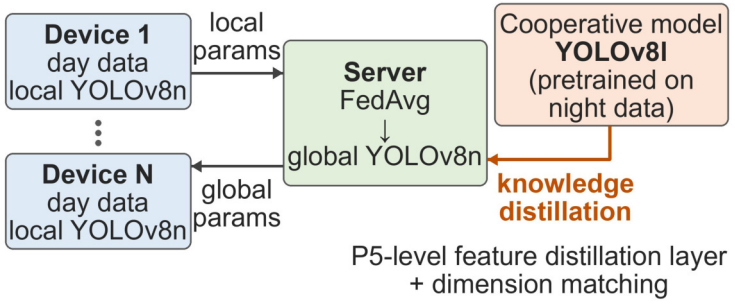
Task stream and operating-day episode structure



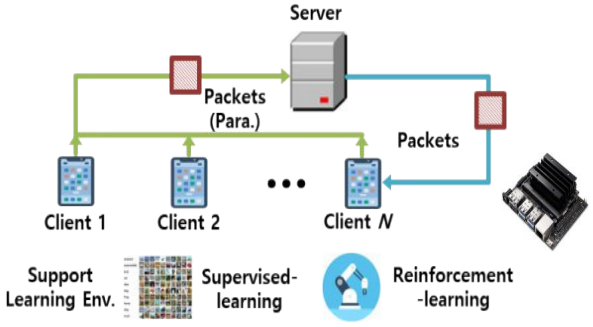
AgentV2G architecture: three role agents on a fixed graph

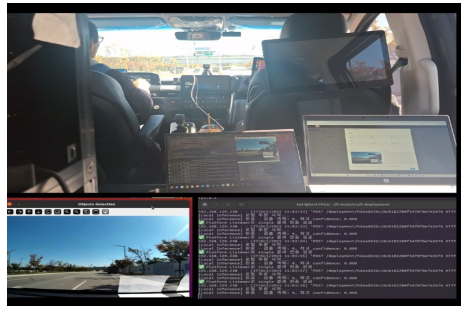
[Federated Learning, Knowledge Distillation] Federated Edge Inference

- **Overview** — Federated inference/learning framework across data-center and edge NPUs (neural processing units) · MSIT / IITP (Institute for Information & Communications Technology Planning & Evaluation), SW Computing Industry R&D (Apr. 2021 – Jun. 2025) · Delivered to Acryl Inc. (the consortium's industry partner) · Code released on GitHub at IITP's request
- **Role: Leader** — problem definition, algorithm design, implementation, experiments, testbed
- **FL (federated learning) testbed built on the open-source Flower framework** (right figure (a)) — The client side was extended with federated RL and per-client non-IID (non-independent, non-identically distributed) data composition; the learning-rate schedule, which Flower adjusts only per epoch, was made adjustable per round and set to cosine annealing; image-transform augmentation was added
  - **Result** — The framework reproduces Google's FedAvg accuracy on CIFAR-10; the schedule and augmentation raise accuracy from 82% to 84%, and auxiliary open data (STL-10) to 86.1%; federated RL was verified on a 2D-navigation task
- **Cooperative federated learning with knowledge distillation** (left figure) — A global YOLOv8n federated from the devices' daytime traffic-sign data generalizes poorly to night, and night data is assumed scarce on the devices. After each FedAvg aggregation, the server distills a cooperative large model (a YOLOv8l pretrained on night traffic-sign data) into the global YOLOv8n through a distillation layer on the backbone's P5-level features with a dimension-matching projection (the distillation layer was developed jointly with Acryl). Night images were synthesized with OpenCV in easy and hard modes from the daytime set that KETI (Korea Electronics Technology Institute) provided
  - **Result** — Night-image accuracy reaches 80.1% (easy) and 36.6% (hard), 20 and 16 points above federated learning alone
- **Edge–server collaborative inference pipeline for real-time traffic sign recognition** (2025; right figure (b): the partners' in-vehicle field test at Hwaseong, Gyeonggi) — The YOLO models and the inference system were deployed on Acryl's Jonathan server and the KETI NPU server
  - The distilled YOLOv8n on the FPGA-based NPU (FPGA: field-programmable gate array) processes routine frames in real time. When a frame's confidence falls below 0.3 (e.g., at night or in rain), the next frame is forwarded to the server's YOLOv8l for federated inference, improving the detection of damaged or occluded signs on difficult frames



Cooperative Federated Learning with YOLO Models

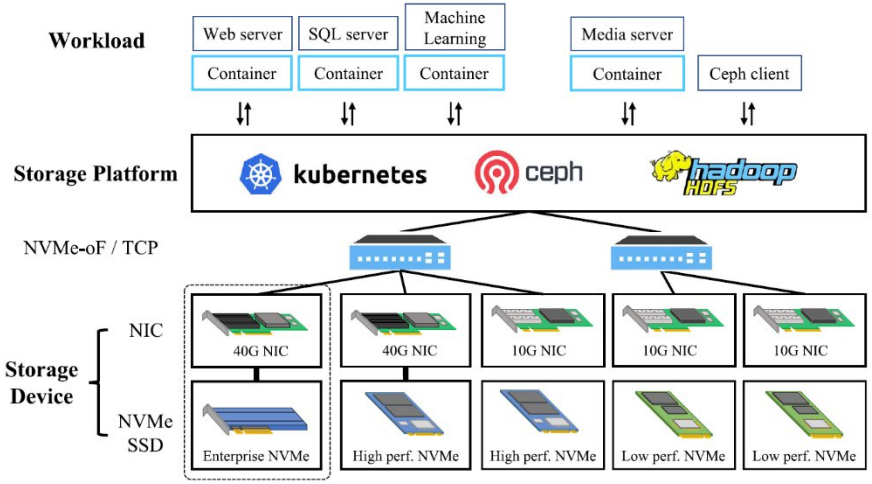




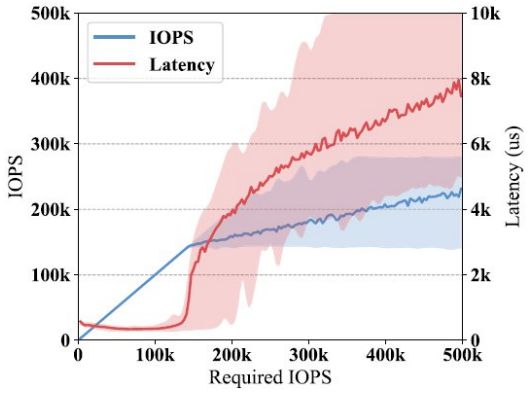
NPU project: (a) embedded FL testbed · (b) in-vehicle field test

[Supervised Learning, Regression] NVMe-oF Storage Simulator with a DNN Surrogate

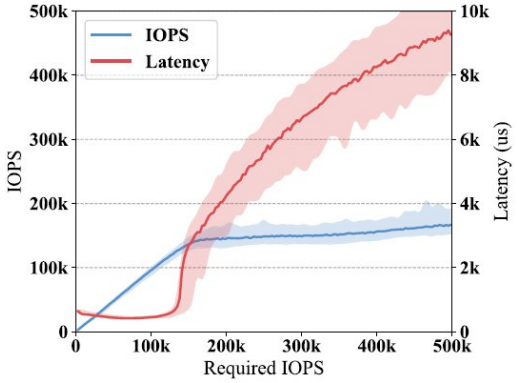
- **Overview** — 2019–2021 · Samsung Electronics project (Aug. 2019 – Jul. 2022) · patents with Samsung: US 12,650,769 B2 (simulation), US 12,210,760 B2 (migration)
- **Role: Leader** — problem definition, algorithm design, experiments
- **Purpose** — To reproduce a disaggregated NVMe-oF (NVMe over Fabrics; here NVMe/TCP) cluster in software (NS-3-based) so that storage-management policies (RL policies included) can be tuned for mixed configurations of heterogeneous devices in a data center
- **Surrogate** — FIO (flexible I/O tester) measures per-node serviced IOPS (I/O operations per second) and latency against required IOPS. A single DNN (deep neural network) surrogate spanning device specifications reproduces both responses (right figure: (a) measured vs. (b) estimated)
  - Four configurations were profiled, from an RC500 SSD on 10 GbE to a Samsung PM1725a SSD on a 40 GbE NIC (network interface card). The surrogate takes a 15-bin histogram of the objects' required IOPS (1,000-IOPS bins) and is trained with an L-infinity loss on serviced IOPS and an L1 loss on latency, and its predicted responses are consistent with the FIO measurements
- **Configurable environment** — Devices are organized under a storage platform that aggregates them in a two-level hierarchy, so that the environment is reconfigured from a system specification rather than rebuilt (left figure)
  - A tier-level object-migration policy is evaluated across homogeneous and heterogeneous compositions of the measured devices under FIO-generated workloads (the scope covered by the migration-control patent)



NVMe-oF disaggregated storage systems



(a) Measured performance pattern

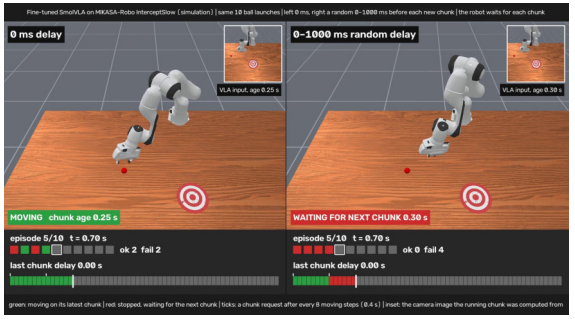


(b) Estimated performance pattern

A surrogate model example for NVMe-oF storage

[RFM, VLA] Off-Board Robot Foundation Models — Serving Control Under Network Delay

- **Overview** — Ongoing research (2026 – present) · follow-on to XferQoE · Collaborative-Intelligence Physical-AI SW-Platform project (MSIT / NIPA, Aug. 2026 – Dec. 2030, researcher)
- **Role: Leader** — problem definition, experiments (research direction, experiment design, and analysis)
- **Target system** — A robot foundation model (RFM) that a home or service robot cannot run at the control rate on board is served from an edge or cloud server. The observation uplink, the inference budget, and the action downlink then enter the control loop, and an action-chunking policy (π0.5 or SmolVLA, chunk length 50) executes several steps from one observation while the scene changes
- **Evidence (measured; rates in table)** — Delay costs success or time: on static tabletop tasks, asynchronous execution (switching to the new chunk, no correction) lost success at 500 ms; synchronous execution (waiting for the new chunk) kept success but cut task throughput (successes per minute) by 35%. On a moving-ball task, both lost success and synchronous fell to 1.0% at 500 ms. Variable delay redrawn after the first chunk (uniform 0–500 or 0–1000 ms) kept the ordering of the two modes and did not differ significantly from fixed delay of equal mean
- **Prior work** — On the policy side, real-time chunking (RTC, NeurIPS 2025), REMAC (ICLR 2026), and Legato (RSS 2026) correct asynchronous chunk execution for inference delay; on the network side, ComVLA (GLOBECOM 2026) fits the visual-token budget of a VLA (vision-language-action model) to the channel capacity and treats transmission delay as negligible. The runs above use neither correction nor token control
- **Proposed work** — Extend XferQoE to the RFM serving path: ① pretrain an RL serving controller in S-Sim on QoS components that do not depend on the RFM, transfer it to the real link through the same three network environments as on page 6, and adapt it to the served RFM by refitting only its objective weights to the robot's task outcomes; upstream of the remote RFM it reads the link state (bandwidth, jitter, queue) and jointly sets observation size, inference budget, and request interval; ② train a lightweight module on the frozen RFM, supervised or by RL, to report each chunk's valid length (actions still usable under delay) and uncertainty, to be adapted alongside the controller. A possible extension is to fine-tune the RFM itself with policy-gradient RL (e.g., RLOO or GRPO) on task-success rewards gathered under network delay



Moving ball: 0 ms vs. 0–1000 ms delay, sync

sonny0714.github.io/demo_video_robot/moving-ball-delay.html

Success rate without correction (LIBERO 100 · MIKASA 200 episodes)

Task and execution	No delay	250 ms fixed	0–500 ms variable	500 ms fixed	0–1000 ms variable
Static tabletop (LIBERO) · async	97%	89%	88%	80%*	88%
Static tabletop (LIBERO) · sync	97%	97%	96%	93%	97%
Moving ball (MIKASA) · async	55.5%	41.0%*	45.5%	32.0%*	38.0%*
Moving ball (MIKASA) · sync	55.5%	7.5%*	10.5%*	1.0%*	3.0%*

* $p < 0.01$ vs. no delay, paired McNemar test · variable = uniform random delay, redrawn per request · LIBERO: π0.5, replan every 20 steps · MIKASA: fine-tuned SmolVLA, replan every 8 steps

Summary and Contact

- **Contact** — Youngseok Lee · yslee.gs@gmail.com · scholar.google.com/citations?user=GhwtxtgAAAAJ

① **Reinforcement Learning** — RL policies for resource management and cloud robotics: a meta-RL rate controller and a bandwidth-allocation policy that transfer to new objectives and environments with little or no retraining (XferQoE · Repot), and a SAC dispatch policy for V2G fleets (AgentV2G)

② **Agentic AI** — AgentV2G uses LLM agents to turn market rules into solver-ready objectives and constraints for V2G operation, with a SAC policy for real-time dispatch; LDuS uses an LLM to turn user requirements into executable loss-function code and validate code and trajectories in a feedback loop for skill adaptation

③ **Systems Engineering** — Both approaches are implemented and validated on purpose-built simulators, surrogate models, and testbeds; five systems built with the same engineering were delivered to Samsung Electronics ×2, Polaris AI, Acryl, and ADD

Selected publications and patents (links)

Type	Title	Venue · Registration	Link
Paper · top-tier	XferQoE: QoS-to-QoE transfer for adaptive rate control	IEEE INFOCOM 2027 · first author under review	—
Paper · top-tier	LLM-based skill diffusion for zero-shot policy adaptation (LDuS)	NeurIPS 2024 Main Conference Track · second author	NeurIPS
Paper · SCIE	Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments	IEEE Access, vol. 9, 2021 · first author	DOI
Paper · SCIE	Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications	IEEE Access, vol. 9, 2021 · second author	DOI
Paper · SCIE	Continual prediction of bug-fix time using deep learning-based activity stream embedding (DASENet)	IEEE Access, vol. 8, 2020 · first author 33 citations	DOI
Patent · first inventor	Graph Embedding-Based Virtual Network Mapping Method	KR 10-2629731 (2024) US 12,047,250 B2 (2024)	KR · US
Patent · first inventor	Network Monitoring System	KR 10-2574070 (2023) US 12,556,492 B2 (2026)	KR · US
Patent · with Samsung	Object Storage System, Migration Control Device, and Migration Control Method	US 12,210,760 B2 (2025)	US
Patent · with Samsung	Simulation Method and Modeling Method	US 12,650,769 B2 (2026)	USPTO
Patent · with Samsung	Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof	US 10,742,742 B2 (2020)	US



성 명	이영석 (Youngseok Lee)	생년월일	
연락처	yslee.gs@gmail.com		
학 력	성균관대학교 전자전기컴퓨터공학과 박사학위 취득예정 (2027.02)		
Google Scholar	scholar.google.com/citations?user=GhwtxtgAAAAJ		
병 역	전문연구요원 복무만료 (2017.03 – 2020.03)		
주 소			

전문분야 (Research Coverage) — 목적·환경 변화(애플리케이션 목적 및 실행 환경)에도 동작을 유지하는 자원 관리 등 closed-loop 제어 시스템을 개발하며, 이를 위해 강화학습 정책을 학습·전이하고 LLM 에이전트 기반 운영 자동화 모델을 구축함. 전용 시뮬레이터·서로게이트·테스트베드를 구축해 두 접근 방법을 검증하고, 시스템 5건을 전달함

① Reinforcement Learning — Resource Management & Cloud Robotics

: 강화학습으로 전송률 제어·대역 할당 등 자원 관리 정책을 학습하고, 애플리케이션의 평가 품질(QoE)·환경이 변화해도 재학습 없이 또는 소량 데이터로 전이하는 알고리즘을 개발해 시뮬레이터·실기기 테스트베드에서 검증

- **XferQoE**(2024–2026, IEEE INFOCOM 2027 심사 중, 제1저자) — 애플리케이션 정보 없이 네트워크 품질 지표(QoS)만으로 사전학습한 단일 전송률 제어기를 재학습 없이 목적 가중치 적합만으로 애플리케이션별 QoE 최적화에 적응시킴
  - ✧ 정책 추론은 엣지 서버가 수행하고 관측만 실제 링크를 건너는 클라우드 로봇 구조에서, 이종 네트워크 환경(실기기 테스트베드 포함)에 사전학습 제어기를 전이하여 QoE를 평가함 (사용 로봇 시뮬레이터: Meta-World·ManiSkill3)
- **Repot**(2020–2021, IEEE Access 2021, 제1저자 · 제1발명자 특허 한·미 등록) — 센싱 디바이스들이 클라우드 서버에 업데이트하는 정보의 품질을 최대로 유지하기 위해, 네트워크 환경 변화에 적응하도록 대역폭 할당 제어를 소규모 데이터로 재학습함
- **원격 로봇 파운데이션 모델(RFM) 서빙 제어**(XferQoE 후속 연구) — 이기종 협업-피지컬AI 과제(2026–2030)와 연계, 엣지 · 클라우드의 RFM이 네트워크 지연 아래 실기기를 구동하는 구조의 서빙 제어기를 설계 중임

② Agentic AI — LLM Agents for System Operation (Robotics & V2G)

: 규칙·요구조건을 실행 가능한 목적식·코드로 변환하고 피드백 루프로 검증하는 LLM 에이전트 구축(로봇 조작 · V2G 운영)

- **LDuS**(2023–2024, NeurIPS 2024, 제2저자) — 학습된 skill diffusion 정책을 문맥·요구조건에 적응시키는 zero-shot 정책 적응 프레임워크로, 정책 전이와 LLM 자동화 두 축을 함께 다룸
  - ✧ 언어로 주어진 미학습 제약(속도·에너지·공간 등)을 LLM이 손실 함수 코드로 변환해 궤적 생성을 유도하고, 생성된 코드와 궤적을 LLM이 피드백 루프로 검증해 잘못된 코드는 재생성하고 궤적이 어긋나면 유도 적용 빈도를 높임
- **AgentV2G**(2026.03–2027.03) — 현대자동차·기아 V2X 과제 실무책임자로 개발한 LLM 에이전트 기반 V2G 플릿 운영 프레임워크로, 전일 입찰부터 실시간 급전 이행까지 이중 정산 전력시장의 전 과정을 단일 파이프라인으로 운용함
  - ✧ 시장 규칙에서 목적식·제약을 도출하는 에이전트, 충전 허브가 관측하지 못하는 상태를 추정하는 에이전트, 최적화 엔진(예: 혼합 정수 선형 계획법) 위에서 모든 입찰·스케줄을 산출하는 에이전트로 세 가지 역할을 분담하게 함

③ Systems Engineering — Simulators, Testbeds, Protocols & Deployment

: 학습 모델을 배포 환경에 맞게 설계·학습하고, 모델 학습·검증용 시뮬레이터·테스트베드와 프로토콜·서비스를 구축함

- 학습 모델 구축 — 엣지 NPU 객체 인식 모델: 주간 표지판 데이터로 연합학습(FedAvg)한 YOLOv8n에 야간 표지판 데이터로 사전학습한 YOLOv8l을 지식증류(backbone P5 증류 층, 아크릴 공동)해 야간 정확도를 연합학습만 대비 최대 20%p 높이고, FPGA NPU 실시간 추론과 저확신 시 다음 프레임 서버 이관 협업 추론으로 구동 · NVMe-oF 성능 서로게이트: FIO 실측으로 학습한 DNN이 노드별 IOPS·지연을 동시에 예측하며 장치 구성 4종에서 실측과 정합
- 시스템 구축 — 네트워크 시뮬레이터·테스트베드: 패킷 수준 C++ 시뮬레이터·NS-3·802.11ac 16노드 테스트베드를 단일 인터페이스로 묶어 제어 정책을 변경 없이 실기기까지 검증 · 삼성전자 전달 2건: 4K 60 Hz 무선 스트리밍용 TCP·신뢰성 UDP 송수신 시스템, 시스템 사양만으로 재구성되는 NVMe-oF 스토리지 시뮬레이터(미국 특허 2건) · IoT 프로토콜: 정의·명세, 고정 크기 자원 식별자 패킷 재설계(삼성전자 공동 특허) · PS-LTE 망 선택: 실측 20만 건 기반 대역 선택 API·앱(Polaris AI 전달)

기술 역량

Languages — C/C++, Python, MATLAB
Machine Learning — JAX/Flax, PyTorch, reinforcement learning (meta-RL, offline RL, in-context adaptation, weight-conditioned multi-objective control, successor features), domain adaptation, representation learning (variational sequence encoder–decoder, graph embedding, Transformer encoders), time-series/world-model prediction, diffusion policies, knowledge distillation, federated learning (Flower, FedAvg)
LLM & Agentic Systems — role-agent orchestration, code-as-policies, binding formulations to solvers, critic-based closed-loop verification
Simulation & Environments — custom packet-level network simulators, vehicular mobility simulation (SUMO), V2G fleet-operation environments, DNN surrogate modeling of measured systems (NVMe-oF storage, FIO benchmarking)
Cloud Robotics — networked robot control (Meta-World · ManiSkill3), off-board policy inference, robot-task QoE modeling, RFM fine-tuning, RFM serving under network delay, VLA fine-tuning (π0.5, SmolVLA), MAE-pretrained ViT encoders
Networking — TCP/UDP protocol implementation (NS-3), IEEE 802.11 family, LTE / PS-LTE, qdisc queue modeling
Systems & Deployment — shared-memory IPC, multithreaded C/C++, Raspberry Pi deployment, YOLO detection on FPGA-based edge NPUs

학력사항 및 경력사항

기간	학교 및 전공	비고
2013.03 – 2027.02 (학위취득예정)	성균관대학교 대학원 전자전기컴퓨터공학과 석박사통합과정 박사학위논문: Reinforcement Learning for Quality-Driven Network Control	GPA 4.33 / 4.5 · 박사과정 수료 2017.02 지도교수: 우홍욱 · CSI-Agent Group
2017.03 – 2020.03	성균관대학교, Software Convergence Research Institute	전문연구요원, 3년 의무복무 만료
2008.03 – 2013.02	성균관대학교 전자전기공학부 (공학사)	GPA 4.23 / 4.5

주요 산학 · 정부 연구과제

기간	과제명	발주기관	역할
2026.03 – 2027.03	V2X 운영 최적화를 위한 강화학습 기반 기술 개발	현대자동차 · 기아	실무책임자
2025.09 – 2025.12	피지컬AI 핵심기술 PoC — 이기종 협업-피지컬AI SDF 특화 기반 모델 연구개발	과기정통부 · NIPA	참여연구원
2024.08 – 2024.12	재난망·철도망 전파간섭 지역 AI 기반 PS-LTE 단말 접속 개선 연구	Polaris AI	실무책임자
2021.04 – 2025.06	데이터센터-엣지 NPU 연합 추론/학습 SW 프레임워크 개발	과기정통부 · IITP	실무책임자
2019.08 – 2022.07	차세대 컴퓨팅환경을 위한 스토리지시스템 및 응용프레임워크 최적화 기술 연구	삼성전자	실무책임자
2017.09 – 2018.09	저 지연 무선 네트워크 프로토콜 개발	삼성전자	실무책임자

이 외 국방과학연구소 · 과학기술정보통신부(구 미래창조과학부) · 교육부 국가 R&D 과제 14건 수행 (연구책임자 1 · 실무책임자 5 · 참여연구원 8)

연구실적 요약

특허 등록 8건 (국내 3 · 미국 5) · 제1발명자 4건 · 삼성전자 공동 등록 3건

시스템 개발 4개 기관에 5건 전달 — 삼성전자(×2) · Polaris AI · Acryl · 국방과학연구소, 전 건 실무책임자

논문 39편 + 심사 중 1편 — SCIE 국제저널 3편 · NeurIPS 2024 Main Track 1편(제2저자) · IEEE INFOCOM 2027 심사 중 1편(제1저자)
(기타 국내외 논문 35편: B급 국제저널 3 · A급 국내저널 5 · B급 국제학술대회 18 · 국내학술대회 9, Google Scholar 참조)

대표 논문

[심사 중] Youngseok Lee and Honguk Woo, "XferQoE: QoS-to-QoE transfer for adaptive rate control," submitted to Proc. IEEE INFOCOM, 2027.

[NeurIPS] Woo Kyung Kim, **Youngseok Lee**, Jooyoung Kim, and Honguk Woo, "LLM-based skill diffusion for zero-shot policy adaptation," Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Main Conference Track.

[SCIE] Youngseok Lee, Woo Kyung Kim, Sung Hyun Choi, Ikjun Yeom, and Honguk Woo, "Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments," IEEE Access, vol. 9, 2021.

[SCIE] Minjae Park, **Youngseok Lee**, Ikjun Yeom, and Honguk Woo, "Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications," IEEE Access, vol. 9, 2021.

[SCIE] Youngseok Lee, Suin Lee, Chan-Gun Lee, Ikjun Yeom, and Honguk Woo, "Continual prediction of bug-fix time using deep learning-based activity stream embedding," IEEE Access, vol. 8, 2020. — 인용 33회

등록 특허 (8건)

등록번호	발명의 명칭	비고
KR 10-2629731 US 12,047,250 B2	Graph Embedding-Based Virtual Network Mapping Method	제1발명자, 한·미 등록
KR 10-2574070 US 12,556,492 B2	Network Monitoring System	제1발명자, 한·미 등록
US 12,210,760 B2	Object Storage System, Migration Control Device, and Migration Control Method	삼성전자 공동 등록
US 12,650,769 B2	Simulation Method and Modeling Method	삼성전자 공동 등록
US 10,742,742 B2	Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof	삼성전자 공동 등록
KR 10-1305307	Simulation Apparatus and Method for Time Synchronization in Spread Spectrum System	GNSS 신호동기 연구

장학 및 수상

기간 / 연도	내역	수여기관
2013.03 – 2018.02	Global Ph.D. Fellowship — 전 기간(5년) 수혜, 2013년 전국 200명 선발	교육부 · 한국연구재단
2008.03 – 2013.02	국가우수장학금(이공계) — 8학기 전액	교육과학기술부 · 한국과학재단
2014, 2015	Outstanding Research Achievement, Global Ph.D. Fellowship Program	교육부 · 한국연구재단
2012.12	IEEE Seoul Section Student Paper Contest — Bronze Award	IEEE Seoul Section

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RESEARCH SUMMARY

- **Achievements**
  - **8 granted patents** (US 5 / KR 3; first inventor on 4) and 9 registered software copyrights
  - **5 systems delivered** to 4 organizations (Samsung Electronics ×2, Polaris AI, Acryl, and the Agency for Defense Development), each as technical leader
  - **39 publications** (including 35 earlier works, 2012–2018) — 3 SCIE journal papers (IEEE Access) and NeurIPS 2024 Main Conference Track (second author); a first-author IEEE INFOCOM 2027 submission is under review
- **Research Coverage** — The work builds resource management and other closed-loop control systems robust to objective and environment shift (application objective and execution environment). Two approaches provide that robustness: RL policy transfer and LLM agents for system operation. Both are implemented and validated on purpose-built simulators, surrogates, and testbeds; five systems were delivered (Systems Engineering).

Research Map

Research area	Representative work	Outcomes
① Reinforcement Learning — Resource Management & Cloud Robotics	○ XferQoE — One QoS-pretrained meta-RL rate controller, adapted per application by objective-weight fitting ○ Repot — Bandwidth allocation transferred across network environments by flow embedding and action shaping ○ RFM (robot foundation model) encoder fine-tuning and real-arm evaluation (π0.5 and SmolVLA on SO-101) ○ Off-board RFMs under network delay — delay-injection evaluation harness (XferQoE follow-up, in progress) ○ Skill-based Code-as-Policies demo for a Physical-AI PoC	○ IEEE INFOCOM 2027 (first author, under review) ○ IEEE Access 2021 (first author) ○ 1 first-inventor patent family (KR·US)
② Agentic AI — LLM Agents for System Operation (Robotics & V2G)	○ LDuS — LLM-guided skill diffusion for zero-shot policy adaptation ○ AgentV2G (in progress) — LLM-agent V2G fleet operation under drift, day-ahead → real-time as one pipeline; SAC policy for real-time dispatch	○ NeurIPS 2024 (second author)
③ Systems Engineering — Simulators, Testbeds, Protocols & Deployment	○ Packet-level C++ training simulator (S-Sim) → NS-3 → 16-node Raspberry Pi NiL testbed with an edge server ○ Code-as-Policies demo pipeline and SO-101 real-arm RFM fine-tuning setup ○ V2G fleet-operation environment — market rules, system marginal price (SMP) series, and EV charging-session records ○ Federated learning with knowledge distillation → YOLOv8n edge-NPU inference with a server-side YOLOv8l ○ NVMe-over-TCP storage simulator with dual-output DNN surrogate ○ LTE band-selection service · QLED wireless streaming testbed ○ IoT network protocol — specification; fixed-size identifier packets ○ Delay-injection evaluation harness for off-board RFMs	○ Delivered ×5 (Samsung ×2, Polaris AI, Acryl, ADD) ○ 3 US patents with Samsung (storage ×2 · IoT identifier packets)
Deep Learning Models (2019–2021)	○ Gemma — Data-center and ISP virtual network mapping learned end to end in one framework ○ DASENet — Bug-fix time re-estimated continually from developer activity streams	○ IEEE Access 2021 (second author) · 2020 (first author) ○ 1 first-inventor patent family (KR·US)

TECHNICAL SKILLS

Languages	C/C++, Python, MATLAB
Machine Learning	JAX/Flax (JIT compilation, custom training loops), PyTorch, reinforcement learning (RL), meta-RL, offline RL, in-context adaptation, domain adaptation, weight-conditioned multi-objective control, successor features, representation learning, variational sequence encoder-decoder, graph embedding, time-series prediction, world-model prediction, diffusion policies, Transformer encoders (self-/cross-attention), knowledge distillation, federated learning (Flower, FedAvg)
LLM & Agentic Systems	role-agent orchestration, code-as-policies, binding formulations to solvers, critic-based closed-loop verification

Simulation & Environments	custom packet-level network simulators, vehicular mobility simulation (SUMO), V2G fleet-operation environments, DNN surrogate modeling of measured systems (NVMe-oF storage, FIO benchmarking)
Cloud Robotics	networked robot control (Meta-World · ManiSkill3), off-board policy inference, robot-task QoE modeling, RFM fine-tuning, RFM serving under network delay, VLA fine-tuning (π0.5, SmolVLA), MAE-pretrained ViT encoders
Networking	TCP/UDP protocol implementation (NS-3), IEEE 802.11 family, LTE / PS-LTE, qdisc queue modeling
Systems & Deployment	shared-memory IPC, multithreaded C/C++, NPU/FPGA inference, embedded deployment (Raspberry Pi), YOLO-based object detection on edge NPUs (FPGA-based)

EDUCATION & MILITARY SERVICE

Period	Degree & Institution	Notes
Mar. 2013 – Feb. 2027 (expected)	Integrated M.S.–Ph.D., Electrical & Computer Engineering, Sungkyunkwan University — Dissertation: *Reinforcement Learning for Quality-Driven Network Control*	GPA 4.33 / 4.5 · Ph.D. coursework completed Feb. 2017 · Advisor: Prof. Honguk Woo, CSI-Agent Group (csiagentgroup.com)
Mar. 2017 – Mar. 2020	Military Service — Expert Research Personnel (전문연구요원), Software Convergence Research Institute, Sungkyunkwan University	Full 3-year term completed
Mar. 2008 – Feb. 2013	B.S., Electronic & Electrical Engineering, Sungkyunkwan University	GPA 4.23 / 4.5

DEVELOPMENT EXPERIENCE

Nineteen systems across the three research axes and the 2019–2021 deep-learning model work.

Reinforcement Learning — Resource Management & Cloud Robotics

[Meta-RL, Transformer, SAC] QoS-to-QoE Transfer Framework for Adaptive Rate Control (XferQoE)
2024 – 2026

- **Supported by** Physical-AI Core-Technology PoC (MSIT / NIPA, 2025) · **Related** IEEE INFOCOM 2027, **first author** (under review)
- **Role: Leader** — problem definition, algorithm design, implementation, experiments, testbed.
- **Built** — One rate controller serves each application over the network: pretrained on QoS (quality-of-service) factors alone, it is then adapted by fitting only the application's objective weights.
- **Model** — SAC (soft actor-critic) with one module per meta-training axis: a regime–context encoder (a bidirectional Transformer with mean pooling → regime latent; a per-patch 1-D CNN with a cross-attention readout → step latent, 8-dim each) and factor critics (one Q-head per QoS factor, ensemble of 3) that compose the value linearly under the objective weight w; the actor reads the state, both latents, and w (policy ≈ 0.3 MB, encoder ≤ 3.5 MB). Meta-pretraining samples a task per episode over the bandwidth dynamics and w in S-Sim.
- **Design** — Robot control, video streaming, and networked monitoring score a delivered stream differently, and their quality feedback is costly to collect. Pretraining needs none of it, and adaptation uses a few feedback shots without a gradient update to the policy, because value is composed per QoS factor under the application's weighting.
- **Result** — From a single pretraining run, adapted QoE (quality of experience) in the simulator averages 91% of the QoE of the per-application end-to-end Oracle (a controller trained on that application's own QoE), and 95–102% on the robot-manipulation tasks. The adapted controller improves QoE over the strongest learning-based baseline within four shots, and the transfer holds from the training simulator to NS-3 and a real-device testbed.

[Transformer, SAC, Policy Transfer] Transferable RL System for Quality-Centric Networked Monitoring (Repot)
2020 – 2021

- **Related** IEEE Access (2021), **first author** · KR 10-2574070 · US 12,556,492 B2 (**first inventor**)
- **Role: Leader** — problem definition, algorithm design, implementation, experiments.
- **Built** — Repot allocates bandwidth among sensing devices sharing one link, so a server's continual query stays accurate as the network degrades.
- **Model** — Flow embedding by a Transformer encoder (per-device MLP vectorization + self-attention over all flows) → a trainable allocation function emitting a latent action → a non-trainable shaping function mapping it to per-device bandwidth limits by a distance-weighted softmax. The policy is trained end to end with SAC in the time-slotted source environment, then frozen while a second, small SAC policy (the adjustment function, with its own

critics) is trained in the NS-3 target; the second policy's outputs enter the shaping function with the frozen latent action.

- **Design** — Each device's stream is represented by its own history and its relation to the others, so a single policy transfers across network scales and conditions by updating a small adjustment function alone. Training takes place in an ideal time-slotted environment where samples are inexpensive; the policy then adapts to an NS-3 environment where that experience would be prohibitive to collect, and its QoE holds as the device count grows.
- **Result** — The policy adapts to a target network environment with 6.25% of the samples required for training from scratch.

[π0.5, SmolVLA, ACT, MAE-Pretrained ViT] Robot Foundation Model Encoder Fine-Tuning and Real-Arm Evaluation (SO-101) 2026

- **Supported by** Physical-AI Core-Technology PoC (MSIT / NIPA, Sep. – Dec. 2025, **researcher**) · follow-on work, Mar. – May 2026
- **Role: Leader** — implementation, experiments (encoder pretraining, data collection, fine-tuning, real-arm tests).
- **Built** — Donut pick-and-place data collected on a low-cost 6-DoF arm by leader-arm teleoperation: 104 episodes (45,553 frames), recollected with horizontal grasps after vertical grasps mostly failed. On LeRobot (PyTorch), the vision encoders of the pretrained SmolVLA and π0.5 were fine-tuned with the action expert frozen; ACT (Action Chunking with Transformers), trained on the same episodes, served as the baseline.
- **Model** — ACT: CNN vision backbone + transformer encoder–decoder trained as a conditional VAE; SmolVLA: SmolVLM-2 vision-language backbone + flow-matching action expert; π0.5: PaliGemma-based VLM + flow-matching action expert. Training ran 100K, 50K, and 30K steps, respectively (batch 8, 8, and 1).
- **Design** — The work targets human-to-robot (H2R) transfer, in which human videos guide robot imitation learning, so the vision encoder is the part trained: H2R was first reproduced on public data (two ViT-B/16 encoders pretrained by masked autoencoding, one on human videos and one on manipulation videos, each serving as the ACT backbone and compared offline by L1/MSE against an ImageNet-pretrained encoder), and the real-arm stage applied the same approach to the pretrained RFMs by fine-tuning their vision encoders alone.
- **Result** — On the real arm, 10 trials each at trained (in-distribution) and new (out-of-distribution) positions: ACT 70% / 40%, SmolVLA 90% / 50%, π0.5 80% / 60%; the gaps between policies are one or two trials of ten and the training budgets differ, so no ranking or pretraining effect is claimed.

[VLA (π0.5, SmolVLA), Action Chunking] Delay-Injection Evaluation Harness for Off-Board Robot Foundation Models 2026 – present

- **Supported by** the Collaborative-Intelligence Physical-AI SW-Platform project (MSIT / NIPA, Aug. 2026 – Dec. 2030) · follow-on to XferQoE
- **Role: Leader** — problem definition, experiments (research direction, experiment design, and analysis).
- **Built** — The harness injects fixed or per-request variable delay into chunk requests on LIBERO (static tabletop, π0.5) and MIKASA (moving ball, fine-tuned SmolVLA) and runs each policy asynchronously (switching to the new chunk) or synchronously (waiting for it), 100–200 episodes per condition with a paired McNemar test.
- **Design** — Delay is injected at the chunk request, where the uplink, the inference budget, and the action downlink of the RFM add up, so one delay value covers the serving path; asynchronous and synchronous execution are the two uncorrected modes, and a moving-ball task is added because a static scene may hide the cost of delay.
- **Result** — Without correction, delay costs success or time: at 500 ms, asynchronous execution falls from 97% to 80% on the static task while synchronous execution keeps its success rate but loses 35% of throughput; on the moving-ball task both fall and synchronous execution drops to 1.0%; variable delay does not differ significantly from fixed delay of equal mean.
- **Planned** — An RL serving controller, pretrained in simulation on RFM-independent QoS components and transferred to the real link as in XferQoE, sets observation size, inference budget, and request interval from the link state and is adapted to the served RFM by refitting only its objective weights to task outcomes; a lightweight module on the frozen RFM reports each chunk's valid length and uncertainty, to be adapted with the controller; a possible extension fine-tunes the RFM by policy-gradient RL (RLOO or GRPO) on task-success rewards under delay.

[VLM (Gemini), LeRobot] Skill-Based Code-as-Policies Demo for a Physical-AI PoC (SO-101 Arms and a LeKiwi Base) 2025 – 2026

- **Supported by** Physical-AI Core-Technology PoC (MSIT / NIPA, Sep. – Dec. 2025, **researcher**)
- **Role: Participant** — pipeline structure design, LeRobot runs; the demo was carried out jointly by the lab.
- **Built** — A four-stage demo (pick → move → assemble → inspect) on two SO-101 arms and a LeKiwi mobile base, recorded in Jan. 2026 as the PoC's 41 s evaluation video: the base's arm picks the parts, the base carries them to the assembly table, two fixed SO-101 arms assemble them, and an inspection camera reports pass/fail.

- **Design** — Code-as-Policies: the task sequence is fixed; Gemini, a vision-language model, fills a code skeleton in the prompt with each skill's code offline (over the robot's perception and control primitives) for a skill library; at run time it reads the camera frame and selects the skill, whose code then drives the arms on LeRobot; the base follows the line without Gemini and the inspection is a camera detection, so no learned policy is in the loop.

Agentic AI — LLM Agents for System Operation (Robotics & V2G)

[Skill VAE, Diffusion Decoder, In-Painting] LLM-Guided Skill Diffusion System for Test-Time Policy Adaptation (LDuS) 2023 – 2024

- **Related** NeurIPS 2024 Main Conference Track, **second author**
- **Role: Leader** — problem definition, algorithm design, architecture, experiment design; robot-task validation on Meta-World and multi-stage Meta-World (3 long-horizon goals).
- **Built** — LDuS steers a pretrained skill diffusion policy at test time with loss-guidance code that an LLM writes from a natural-language context (speed, energy, or spatial constraints).
- **Model** — A hierarchical skill model pretrained on goal-annotated expert trajectories: a VAE skill encoder maps an h-step state–action segment to a skill embedding, a skill prior predicts it from the state and the language goal encoded by CLIP (contrastive language–image pretraining), and a skill diffusion planner (a U-Net denoiser as in Diffuser) reconstructs the segment from the embedding. At test time GPT-3.5 writes the loss code from the context, the loss gradient guides every denoising step, and GPT-4 judges contact from the camera frame to gate constraints (such as a force constraint) that apply only while touching the object.
- **Design** — Sequential in-painting conditions target trajectories on past state–action sequences, with no policy retraining. A closed loop validates the generated code and the guided trajectories against the context, regenerating faulty code and raising the guidance frequency until the trajectories satisfy the context.
- **Result** — On Meta-World speed constraints, LDuS reaches a 94.6% success rate and 87.4 context reward, against 76.4% and 69.8 for Diffuser+Guidance (the same gradient guidance with a hand-written loss). On multi-stage Meta-World tasks, it reaches 84.0% and 82.0, against 36.7% and 45.9.

[LLM Agents, RL Solver, MILP, Drift Detection] Agentic Scheduling Framework for V2G Fleet Operation (AgentV2G) 2026 – present

- **Funded by** the Hyundai Motor · Kia V2X project (Mar. 2026 – Mar. 2027, ongoing) · **technical leader**
- **Role: Leader** — problem definition, algorithm design, implementation, experiments; construction of the V2G fleet-operation environment (Jeju two-settlement pilot-market rules).
- **Built** — AgentV2G operates an EV charging hub in a two-settlement electricity market, integrating day-ahead bidding and real-time dispatch in a single pipeline.
- **Model** — One open-weight LLM (Llama-3.1-8B-Instruct) without fine-tuning serves every agent on a static orchestration graph with dynamic tool selection; an orchestrator relays typed artifacts between agents, and the objective is immutable within an episode, so no agent can relax the objective it is evaluated against.
- **Design** — The work is divided among three role agents: one derives the objective and constraints from the market rules, one maintains a belief state over what the hub cannot observe, and one produces every bid and schedule through a mixed-integer linear programming (MILP) engine. The agents are re-invoked by drift metrics when prices, session behavior, or the market rules change, and when an output fails validation, a deterministic fallback per role (last validated formulation, persistence estimate, or proportional allocation) still returns a feasible schedule.
- **RL solver** — For the real-time dispatch stage, a SAC (soft actor-critic) policy was built to replace the MILP engine. A shared MLP embeds each EV's observation, and the mean of these local embeddings forms the fleet embedding. A shared head reads the local and fleet embeddings to emit each EV's charge/discharge ratio, so the policy is independent of fleet size (a set-attention encoder is also implemented). The action is learned as a residual over rule-based dispatch and can be projected onto the allowed band; the Jeju settlement penalties (band deviation, unmet charging) and battery degradation enter the actor loss as a regularization term. Evaluation on 200-EV days (KPX Jeju prices, ACN sessions) compares the policy with MILP, rule-based, and oracle dispatch on a normalized profit metric.
- **Result** — A 10,000-EV day is scheduled in ≈54 s, within Hyundai Motor's 90 s limit (feedback loop disabled).

Systems Engineering — Simulators, Testbeds, Protocols & Deployment

Federated Learning & Edge Deployment

[Federated Learning, Knowledge Distillation, YOLOv8n] Edge–Server Collaborative Inference Pipeline for Real-Time Traffic Sign Recognition 2025

- **Supported by** MSIT / IITP, SW Computing Industry R&D program (Apr. 2021 – Jun. 2025) · **technical leader** (cooperative federated-learning structure, YOLO models, and inference system) · **Delivered to** Acryl Inc., the industry partner in the project consortium
- **Role: Leader** — problem definition, algorithm design, implementation, experiments, testbed.
- **Built** — Two-tier road-sign recognition for road-traffic monitoring, flagging damaged or occluded signs per frame. The edge YOLOv8n was produced by cooperative federated learning: FedAvg over the devices' daytime traffic-sign data, then server-side knowledge distillation from a YOLOv8l pretrained on night traffic-sign data through a distillation layer on the backbone's P5-level features with a dimension-matching projection.
- **Model** — YOLOv8n on the edge and YOLOv8l on the server. The Flower-based federated-learning framework was extended on the client side with federated RL and per-client non-IID (non-independent, non-identically distributed) data composition; Flower's per-epoch learning-rate schedule was made adjustable per round and set to cosine annealing, and image-transform augmentation was added; the schedule and augmentation raise CIFAR-10 accuracy from 82% to 84% (86.1% with STL-10 auxiliary data) on the framework's supervised benchmark.
- **Design** — The distilled YOLOv8n on the FPGA-based NPU processes routine frames in real time. When a frame's confidence falls below 0.3 (e.g., at night or in rain), the next frame is forwarded to the larger server-side model for federated inference, improving accuracy on the hard frames without adding edge latency to the easy ones. Night driving data is assumed to be scarce on the devices, so night is treated as the unseen condition: night images were synthesized from the daytime set with OpenCV in easy and hard modes, and the night-pretrained large model supplies the knowledge the federated model lacks, without moving device data.
- **Result** — Night-image accuracy reaches 80.1% (easy) and 36.6% (hard), 20 and 16 points above federated learning alone; the in-vehicle field test at Hwaseong was run by the consortium partners.

Simulators, Testbeds & Surrogates

[C++ Packet Simulation, NS-3, 802.11ac] Network Environments of Increasing Fidelity for Rate-Control Policy Pretraining and Validation (S-Sim, NS-3, NiL Testbed) 2020 – 2026

- **Role: Leader** — implementation, experiments, testbed.
- **Built** — XferQoE's training backend and validation path: a custom µs-resolution packet-level C++ simulator with no protocol stack (S-Sim), a full-stack NS-3 environment, and a Raspberry Pi network-in-the-loop (NiL) testbed (2026) that puts a real wireless link in the control loop. On the testbed, Raspberry Pi 4 senders share one IEEE 802.11ac link through an access point to an edge server (one sender for robot control, up to 16 for monitoring).
- **Design** — The three environments share one state-input, rate-output interface and replay the same bandwidth trace (Puffer), so a policy pretrained entirely in simulation runs on all three unchanged; S-Sim adds parallel multi-environment rollout and replay of eight bandwidth traces (four synthetic, four real-world). On the testbed, C/C++ TCP/UDP daemons implement episode-level clock synchronization with drift measurement, sender-side virtual qdisc queue modeling, and frame-deadline expiry accounting; the last two track the traffic.
- **Result** — On the testbed, policy inference and rate computation take 6.7 ms of each 1 s control interval.

[NS-3, FIO] NVMe-over-TCP Disaggregated Storage Simulator with a Dual-Output DNN Surrogate 2019 – 2021

- **Funded by** Samsung Electronics (Aug. 2019 – Jul. 2022) · **technical leader** · **Delivered to** Samsung Electronics · **Related** US 12,210,760 B2 · US 12,650,769 B2 · SW Reg. C-2020-037999
- **Role: Leader** — problem definition, algorithm design, experiments.
- **Built** — The simulator reproduces a disaggregated NVMe-over-TCP cluster in software, so storage management policies can be tuned for the mixed device generations and configurations a production data center runs.
- **Model** — The surrogate takes a 15-bin histogram of the objects' required IOPS (1,000-IOPS bins) and predicts serviced IOPS and latency with one DNN, trained with an L-infinity loss on serviced IOPS and an L1 loss on latency over FIO logs in which each object's required IOPS is set at random.
- **Design** — FIO measures per-node serviced IOPS and latency against required IOPS, and one DNN surrogate spanning device specifications reproduces both responses. Devices are organized under a platform that aggregates them in a two-level hierarchy, so the environment is reconfigured from a system specification rather than rebuilt. A tier-level object-migration policy is evaluated across homogeneous and heterogeneous compositions of the measured devices under FIO-generated workloads.

- **Result** — Predicted IOPS and latency responses are consistent with FIO measurements on the four profiled NVMe-over-TCP configurations, from an RC500 SSD on 10 GbE to a Samsung PM1725a SSD on a 40 GbE NIC.

[TCP/UDP, Raw-Socket Sniffer, 802.11ac] Low-Latency Wireless Streaming Testbed for the Samsung QLED TV One Connect System 2018

- **Funded by** Samsung Electronics (Sep. 2017 – Sep. 2018) · **technical leader** · **Delivered to** Samsung Electronics
- **Role: Leader** — implementation, experiments, testbed.
- **Built** — The testbed reproduces the Samsung One Connect link that feeds a QLED TV; transmitter and receiver both run on Linux hosts with Broadcom BCM4360 wireless NICs.
- **Design** — Layer-by-layer delay analysis of the standard stack for 4K 60 Hz video. Three C/C++ components generate and measure the traffic: a TCP sender–receiver with buffer-based flow control, a UDP sender–receiver with bitrate-paced transmission, and a raw-socket sniffer measuring RTT, throughput, packet loss, duplication, and reordering.
- **Result** — Bitrate-paced UDP transmission sustains 4K 60 Hz frames at 6 ms average RTT (TCP path: 11 ms).

Evaluation Environments & Training Framework

[Meta-World, ManiSkill3, YOLO Scoring] Multi-Application QoE Evaluation Environments for Network Policy Learning 2024 – 2026

- **Supported by** Physical-AI Core-Technology PoC (MSIT / NIPA, 2025)
- **Role: Leader** — implementation, experiments.
- **Built** — Application layer of XferQoE: three downstream environments that consume a network-delivered stream and score it with a quality model of their own.
- **Design** — Where a learned model decides quality, delivery loss and staleness propagate through that model's decision boundary rather than through a designer's formula. Robot control (Meta-World, ManiSkill3), networked monitoring (16 devices scored live by a YOLO detector), and video streaming (live and VoD) each weigh the QoS factors differently.

[SAC, JIT/vmap] Unified JAX/Flax Training Framework for Network Control Policies 2020 – 2026

- **Role: Leader** — implementation.
- **Built** — A shared soft actor-critic (SAC) implementation supports all seven XferQoE baselines and three hand-crafted controllers, so every method is compared under identical conditions.
- **Design** — Encoders, decoders, and actor-critic are organized under one three-tier abstraction (layer / block / module) with a single dataclass I/O convention, so supervised learning and RL share components. Fixing hyperparameters as static arguments compiles the training step into one JIT kernel, and per-task parameter stacks vectorized with vmap process every parallel environment in a single forward pass.

[Gym Environment (EV2Gym), Market-Rule Presets] V2G Fleet-Operation Evaluation Environment for Two-Settlement Markets 2026 – present

- **Role: Leader** — problem definition, implementation.
- **Built** — Charging-hub environment implementing the Jeju two-settlement pilot rules, with EU and NEM (Australian National Electricity Market) presets.
- **Design** — Korea Power Exchange (KPX) SMP series and EV charging-session records drive it. Price, session behavior, and the rules themselves drift alone or together, so a scheduler is scored on episode-level operating quality and on how fast it recovers after the drift on the Jeju pilot-market rules.

Delivered Services & Tools

[PostGIS, HTTP API, Flutter] Database-Backed Network Selection Application for Mobile Terminals 2024

- **Funded by** Polaris AI (Aug. 2024 – Dec. 2024) · **technical leader** · **Delivered to** Polaris AI
- **Role: Leader** — algorithm design, implementation.
- **Built** — The application recommends the network a public-safety LTE (PS-LTE) terminal should use, PS-LTE or a commercial LTE band, in areas where the disaster-response network and the railway network share a band and interfere.
- **Design** — A PostGIS database of over 200,000 normalized field measurements backs the ranking of 6 LTE bands (10 carriers) by RSRP, RSRQ, and SINR, using the nearest measurements in sparsely covered areas. A server provides that ranking and the threshold-based handover decision between PS-LTE and the commercial network via an HTTP API, and a Flutter client presents the recommendation on the terminal.

[MATLAB, Monte Carlo] GUI-Based Analysis Tool for Direct GPS P(Y) Code Acquisition under Wideband Jamming 2013

- **Funded by** ADD Center for Defense Satellite Navigation (SN-42), Mar. 2013 – Oct. 2013 · **technical leader** · **Delivered to** ADD
- **Role: Leader** — implementation, experiments.
- **Built** — MATLAB GUI tool that simulates direct acquisition of the military GPS P(Y) code, bypassing the C/A code; Monte Carlo analysis of mean acquisition time across jamming-to-signal ratios of 0–42 dB gives the maximum jamming level under which acquisition still succeeds.

Protocol & Packet Design

[MQTT, Packet Design] IoT Network Protocol and Packet Structure for an Intelligent Gateway 2014 – 2017

- **Funded by** the MSIP Global Research Program (Nov. 2014 – Jun. 2017) · **technical leader** · **Related** US 10,742,742 B2 · held jointly with Samsung Electronics (**third of four inventors**)
- **Role: Leader** — algorithm design (protocol specification and packet-structure redesign).
- **Built** — The gateway–device network protocol was defined and specified end to end by the author: an MQTT-based message protocol whose packet structure was redesigned around fixed-size sub-resource identifiers.
- **Design** — A controlling apparatus assigns each device one fixed-size identifier item per level of its URI hierarchy; the device stores it, requests and commands carry it, and both ends keep the mapping in a resource access table.
- **Result** — Packets carry the fixed-size identifier instead of the long, variable-length URI that wastes network capacity, so packet size decreases and transfer efficiency improves.

Deep Learning Models

[GCN Encoder, Mapping Decoder, REINFORCE] RL-Based Graph Embedding and Mapping System for Virtual Networks (Gemma) 2020 – 2021

- **Related** IEEE Access (2021), **second author** · KR 10-2629731 · US 12,047,250 B2 (**first inventor**)
- **Role: Participant** — algorithm design (network structure), experiment design.
- **Built** — Gemma maps a requested virtual network onto the physical substrate that will host it; this is the resource-allocation problem behind data-center tenancy, ISP network slicing, and service-function chaining.
- **Model** — A GCN (graph convolutional network) encoder embeds the virtual and the substrate networks. A node-mapping decoder selects the top-ranked virtual node and scores it on each substrate node through two MLP rank functions (two-stage mapping); the encoder and decoder are trained end to end by a modified REINFORCE that maximizes the problem-specific expected reward by gradient ascent.
- **Design** — One learned structure covers both request types without being rebuilt: static (offline batch) requests that hold their allocation once placed, and dynamic (online) ones that arrive over time and are judged on placement speed. Iterative decoder ranking lets the same model handle substrates of varied structure and capacity.
- **Result** — Gemma outperforms heuristic and learning-based methods on data-center online arrivals and ISP offline batch slicing, and re-maps allocations under substrate capacity changes at a tunable reconfiguration cost.

[LSTM, Pretrained Embedding Transfer] Activity-Stream Embedding Model for Continual Bug-Fix Time Prediction (DASENet) 2019 – 2020

- **Related** IEEE Access (2020), **first author**
- **Role: Leader** — problem definition, algorithm design, implementation, experiments.
- **Built** — DASENet predicts how long an open bug will take to fix and revises the estimate as developer activity arrives, so release planning and staffing can be adjusted while the bug is still open.
- **Model** — Each bug report becomes a sequence of per-day activity bins. Per bin, deep BLSTMs (bidirectional LSTMs) encode the user comments and the system records, an MLP encodes the metadata, and a merging MLP yields the bin embedding; a residual LSTM (RLSTM) sequence network over the bin embeddings yields the activity-stream embedding; the whole network is pretrained through an MLP and softmax over bug-fix-time classes. A task MLP stacked on the pretrained network is then trained separately for each task, without end-to-end fine-tuning.
- **Result** — Trained on about 900 samples (2% of the full dataset) on top of the pretrained activity-stream embedding, a variant task model stays competitive with full-data models, showing the learned structure transfers.

PUBLICATIONS

- The five works below are listed in full; 35 further works from 2012 to 2018 (3 other international journals, 5 domestic journals, 18 international conferences, 9 domestic conferences) are on Google Scholar.

- Since 2018, the research has focused on learning-based network control and policy transfer with Prof. Honguk Woo; earlier work was conducted in two other groups at the same university — content-centric networking, IoT, and vehicular communication with Prof. Ikjun Yeom, and GNSS signal synchronization with Prof. Seokho Yoon.

Google Scholar: scholar.google.com/citations?user=GhwtxtgAAAAJ

Top-Tier Conferences

1. [Under review] **Youngseok Lee** and Honguk Woo**,** "XferQoE: QoS-to-QoE transfer for adaptive rate control," submitted to *Proc. IEEE INFOCOM*, 2027.
2. Woo Kyung Kim, **Youngseok Lee**, Jooyoung Kim, and Honguk Woo, "LLM-based skill diffusion for zero-shot policy adaptation," in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, Main Conference Track.

International Journals (SCIE)

1. **Youngseok Lee**, Woo Kyung Kim, Sung Hyun Choi, Ikjun Yeom, and Honguk Woo, "Repot: Transferable reinforcement learning for quality-centric networked monitoring in various environments," *IEEE Access*, vol. 9, pp. 147280–147294, 2021.
2. Minjae Park, **Youngseok Lee**, Ikjun Yeom, and Honguk Woo, "Gemma: Reinforcement learning-based graph embedding and mapping for virtual network applications," *IEEE Access*, vol. 9, pp. 105463–105476, 2021.
3. **Youngseok Lee**, Suin Lee, Chan-Gun Lee, Ikjun Yeom, and Honguk Woo, "Continual prediction of bug-fix time using deep learning-based activity stream embedding," *IEEE Access*, vol. 8, pp. 10503–10515, 2020. **(33 citations)**

PATENTS

Granted: 8 (United States 5, Republic of Korea 3) · First inventor on 4

- The eight granted patents cover six distinct inventions; the two first-inventor inventions are registered in both Korea and the United States, each corresponding to an IEEE Access publication.

Invention	Registration(s)	Granted	Notes
Graph Embedding-Based Virtual Network Mapping Method	KR 10-2629731 · US 12,047,250 B2	KR 2024 · US 2024	First inventor · SKKU · Related: Gemma, IEEE Access (2021)
Network Monitoring System	KR 10-2574070 · US 12,556,492 B2	KR 2023 · US 2026	First inventor · SKKU · Related: Repot, IEEE Access (2021)
Object Storage System, Migration Control Device, and Migration Control Method	US 12,210,760 B2	2025	Jointly held with Samsung Electronics
Simulation Method and Modeling Method	US 12,650,769 B2	2026	Jointly held with Samsung Electronics
Electronic Apparatus, System for Internet-of-Things Environment and Control Method Thereof	US 10,742,742 B2	2020	Jointly held with Samsung Electronics
Simulation Apparatus and Method for Time Synchronization in Spread Spectrum System	KR 10-1305307	2013	SKKU · GNSS signal-synchronization work

SOFTWARE COPYRIGHT REGISTRATIONS

Each program below is registered with the Korea Copyright Commission (registrant: Research & Business Foundation, Sungkyunkwan University); the numbers cited in Development Experience refer to this list.

Reg. No.	Title	Created	Registered
C-2023-048806	Hierarchical Meta-Reinforcement Learning Model	Oct. 19, 2023	Nov. 3, 2023
C-2021-054170	Federated Learning Performance Evaluation for Heterogeneous Datasets (Flower Framework)	Nov. 24, 2021	Dec. 9, 2021
C-2021-041557	Behaviour Analysis Program for Risk-Sensitive Reinforcement Learning	Sep. 16, 2021	Oct. 20, 2021
C-2021-035993	Replay-Buffer Sharing for Accelerated Personalized Federated Reinforcement Learning	Aug. 25, 2021	Sep. 8, 2021
C-2020-039571	SUMO Simulator-Based Clustering Framework	Oct. 15, 2020	Nov. 4, 2020
C-2020-038567	Backscatter Network Simulator for the EPC Gen2 Protocol	Oct. 22, 2020	Oct. 29, 2020
C-2020-037999	ns-3-Based NVMe-oF Disaggregated Storage Simulator for Reinforcement Learning	Sep. 5, 2020	Oct. 27, 2020
C-2016-027920	Signal Synchronization Performance Measurement for Galileo under Noisy Multipath	Oct. 28, 2016	Nov. 18, 2016
C-2016-020663	Cache Performance Measurement Program for Content-Centric Networking	Jul. 14, 2016	Aug. 24, 2016

FUNDED PROJECTS

- **20 funded projects** — Principal investigator 1 · technical leader 10 · researcher 9; sponsors span MSIT/MSIP, MOE, Samsung Electronics, Hyundai Motor · Kia, Polaris AI, and ADD

MSIT = Ministry of Science and ICT; MSIP = its predecessor, the Ministry of Science, ICT and Future Planning (before 2017); MOE = Ministry of Education; NIPA = National IT Industry Promotion Agency; IITP = Institute for Information & Communications Technology Planning & Evaluation. Roles: principal investigator (연구책임자), technical leader (실무책임자), researcher (참여연구원).

Reinforcement Learning — Resource Management & Cloud Robotics

Project	Sponsor / Program	Role	Period
이기종 협업-피지컬 AI SDF 특화 기반 모델 연구 개발	MSIT / NIPA, Collaborative-Intelligence Physical-AI SW-Platform R&D Ecosystem — Hosted by SKKU	Researcher	Aug. 2026 – Dec. 2030 (ongoing)
실세계 범용 체화 지능 실현을 위한 반응형 스킬 디퓨전 연구	MSIT / Core Research (Leap)	Researcher	Mar. 2026 – Feb. 2029 (ongoing)
피지컬 AI 핵심기술 PoC — 이기종 협업-피지컬 AI SDF 특화 기반 모델 연구개발 (제 2 과제)	MSIT / NIPA, Industry-Specific Physical-AI Leading-Model PoC	Researcher	Sep. 2025 – Dec. 2025
지속학습을 통한 인사이트 고속 추론 연구	MSIT / IITP, Human-Centered Next-Gen AI	Researcher	Apr. 2025 – Dec. 2029 (ongoing)
멀티모달 스킬 전이 기반 정책 일반화 기법 연구	MSIT / Young Researcher (Excellent)	Researcher	Mar. 2023 – Feb. 2026
동적인 디바이스 환경에서 적응적 연합학습기술 개발	MSIT / IITP, Next-Gen AI Core	Technical leader	Apr. 2021 – Dec. 2022

Agentic AI — LLM Agents for System Operation (Robotics & V2G)

Project	Sponsor / Program	Role	Period
V2X 운영 최적화를 위한 강화학습 기반 기술 개발	Hyundai Motor · Kia	Technical leader	Mar. 2026 – Mar. 2027 (ongoing)
개성 형성이 가능한 에이전트 플랫폼 기술 개발	MSIT / IITP, Human-Centered AI Core	Researcher	Apr. 2022 – Dec. 2026 (ongoing)
오픈 도메인 멀티모달 자기주도 인공지능 기술 개발	MSIT / IITP, Human-Centered AI Core	Researcher	Apr. 2022 – Dec. 2026 (ongoing)

Systems Engineering — Simulators, Testbeds, Protocols & Deployment

Project	Sponsor / Program	Role	Period
차세대 컴퓨팅환경을 위한 스토리지시스템 및 응용프레임워크 최적화 기술 연구	Samsung Electronics	Technical leader	Aug. 2019 – Jul. 2022
재밍대응 식별에 의한 위성항법신호 복원기법 연구	Agency for Defense Development (SN-42)	Technical leader	Mar. 2013 – Oct. 2013
재난안전통신망과 철도망 전파간섭 지역에서 AI 기반 대규모 PS-LTE 단말기 접속방법 개선에 관한 연구	Polaris AI	Technical leader	Aug. 2024 – Dec. 2024
데이터센터-엣지 NPU 간 연합 추론/학습 기반 대규모 인공지능 응용/개발을 용이하게 하는 SW 프레임워크 개발	MSIT / IITP, SW Computing Industry	Technical leader	Apr. 2021 – Jun. 2025
저 지연 무선 네트워크 프로토콜 개발	Samsung Electronics	Technical leader	Sep. 2017 – Sep. 2018

Early Work — Networking, GNSS, IoT & Media Platforms

Project	Sponsor / Program	Role	Period
소프트웨어 정의 커넥티드 카	MSIP / Individual Research (Strategic)	Technical leader	Dec. 2016 – Nov. 2019
사물인터넷을 위한 지능형 게이트웨이	MSIP / Global Research Program	Technical leader	Nov. 2014 – Jun. 2017
유무선 통합망을 위한 콘텐츠 중심 네트워크 기술	MSIP / Basic Research Program	Technical leader	Mar. 2014 – Apr. 2015
스마트 TV 2.0 소프트웨어 플랫폼	MSIP / ICT & Broadcasting R&D	Researcher	Mar. 2014 – Aug. 2016
BOC 신호 추적을 위한 새로운 상관함수에 대한 연구	MOE / Global Ph.D. Fellowship	Principal investigator	Mar. 2013 – Feb. 2018
초정밀 위치 기반 서비스를 위한 여러 대역 신호 및 이종 항법위성 기반 신호처리기술	MSIP / Core Individual Research	Researcher	Mar. 2013 – Oct. 2013

HONORS & FELLOWSHIPS

Year	Honor / Fellowship	Awarded by
2014 · 2015	Outstanding Research Achievement, Global Ph.D. Fellowship Program	Ministry of Education / NRF Korea
Mar. 2013 – Feb. 2018	Global Ph.D. Fellowship — Full 5-year term; 200 selected nationwide in 2013	Ministry of Education / NRF Korea
Dec. 2012	IEEE Seoul Section Student Paper Contest — Bronze Award	IEEE Seoul Section
Nov. 2012	Honorable Mention, 9th Undergraduate Thesis Presentation Contest	Sungkyunkwan University
Mar. 2008 – Feb. 2013	National Science & Engineering Scholarship — Full tuition, 8 semesters	MEST / Korea Science & Engineering Foundation